

L5: Diameter and Efficiency of $G(n,p)$ Versus Regular Networks

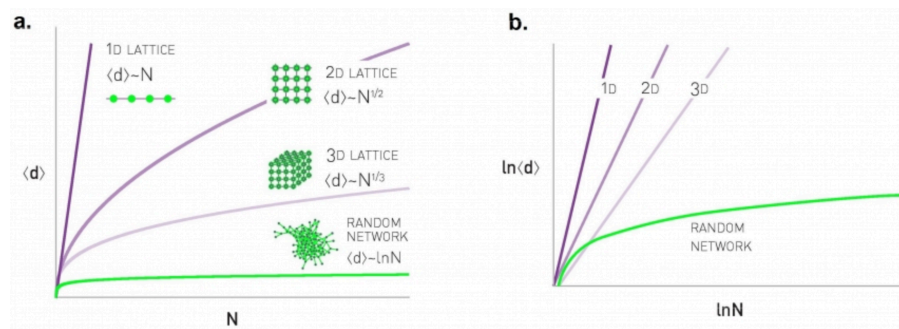
Does the diameter of all networks increase logarithmically? Clearly not. Let's examine what happens in regular networks – and more specifically, in lattice networks.

In one-dimensional lattices, each node has two neighbors, and the lattice is a line network of n nodes -- so the diameter increases linearly with n .

In two dimensions, each node has 4 neighbors, and the lattice is a square with $\sqrt{n}$ nodes on each side -- so the diameter increases as $O(n^{1/2})$

Similarly, in three dimensions each node has 6 neighbors, and the diameter grows as $O(n^{1/3})$ – and so on in higher dimensions.

This suggests that in lattice networks the diameter grows as a power-law of the network size – this grows much faster than a logarithmic function.



Let us go back to the hypothetical social network of 7 billion humans – and with an average of 6 connections per person. If this social network was a regular lattice, it would need to have 3 dimensions so that each node has 6 connections. So the diameter would be in the order of $n^{1/3}$, which is about 1,913 – this is much larger than the diameter we derived earlier for a random network of the same size and density - this being about 22.

These simple derivations show a major qualitative difference between regular networks and random networks: in the former, the diameter and CPL increase much faster than in the latter – as a power-law in the former and logarithmically in the latter.

Food For Thought

Pick a non-lattice regular network and examine if the diameter still increases as a power-law.